

Logistic regression: Calculating a probability with the sigmoid function

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Logistic regression models output probabilities, which can be used directly or converted to binary categories.

The sigmoid function ensures the output of logistic regression is always between 0 and 1, representing a probability.

A logistic regression model uses a linear equation and the sigmoid function to calculate the probability of an event.

The log-odds (z) represent the log of the ratio of probabilities for the two possible outcomes.

Many problems require a probability estimate as output. **Logistic regression** is an extremely efficient mechanism for calculating probabilities. Practically speaking, you can use the returned probability in either of the following two ways:

Applied "as is." For example, if a spam-prediction model takes an email as input and outputs a value of 0.932, this implies a 93.2% probability that the email is spam.

Converted to a [binary category](#) such as True or False, Spam or Not Spam.

This module focuses on using logistic regression model output as-is. In the [Classification module](#), you'll learn how to convert this output into a binary category.

You might be wondering how a logistic regression model can ensure its output represents a probability, always outputting a value between 0 and 1. As it happens, there's a family of functions called **logistic functions** whose output has those same characteristics. The standard logistic function, also known as the [sigmoid function](#) (*sigmoid* means "s-shaped"), has the formula:

$$f(x) = \frac{1}{1 + e^{-x}}$$

where:

$f(x)$ is the output of the sigmoid function.

e is [Euler's number](#): a mathematical constant ≈ 2.71828 .

x is the input to the sigmoid function.

Figure 1 shows the corresponding graph of the sigmoid function.

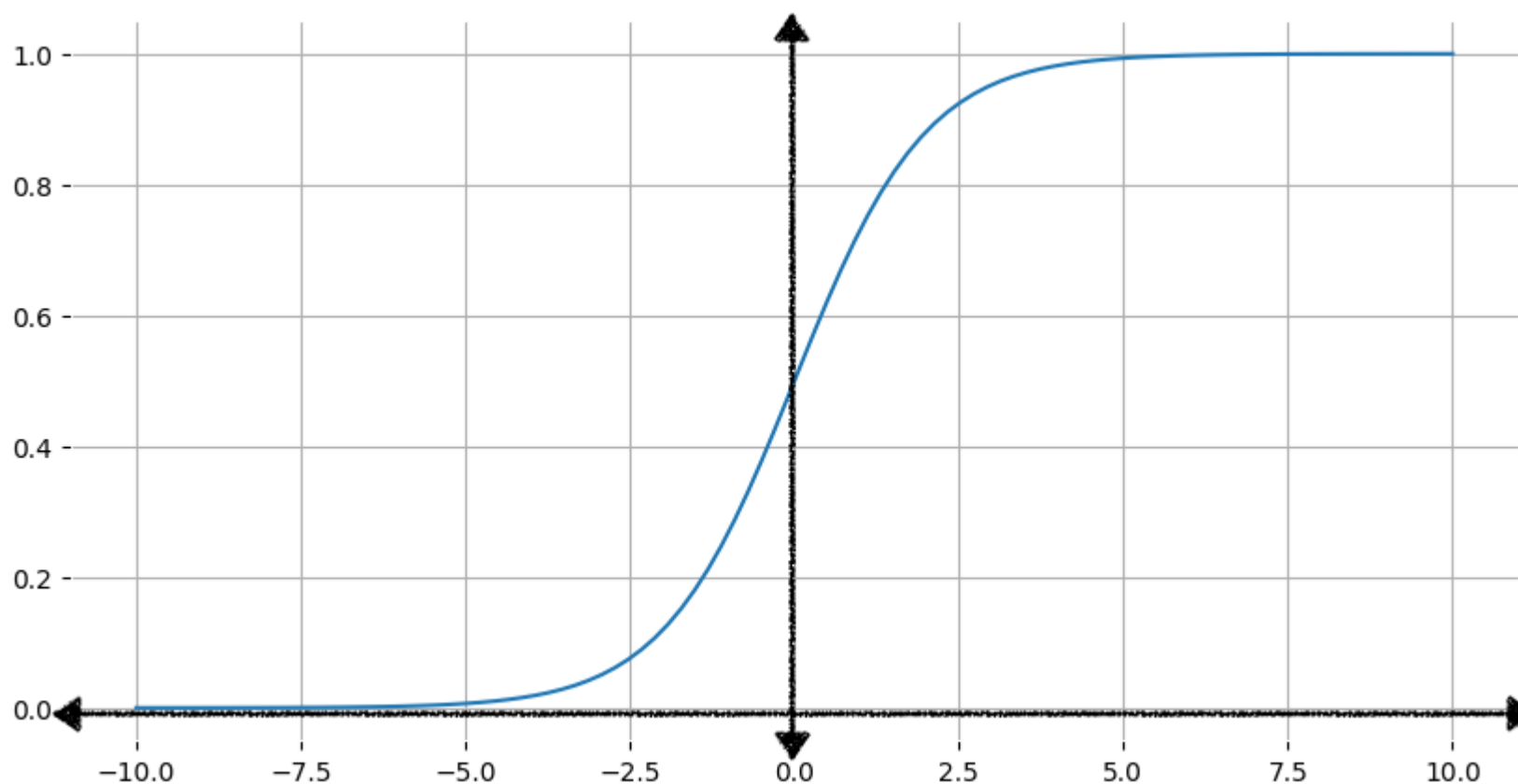


Figure 1. Graph of the sigmoid function. The curve approaches 0 as x values decrease to negative infinity, and 1 as x values increase toward infinity.

As the input, x , increases, the output of the sigmoid function approaches but never reaches 1. Similarly, as the input decreases, the sigmoid function's output approaches but never reaches 0.

Click here for a deeper dive into the math behind the sigmoid function

The table below shows the output values of the sigmoid function for input values in the range -7 to 7 . Note how quickly the sigmoid approaches 0 for decreasing negative input values, and how quickly the sigmoid approaches 1 for increasing positive input values.

However, no matter how large or how small the input value, the output will always be greater than 0 and less than 1.

Input	Sigmoid output
-7	0.001
-6	0.002
-5	0.007
-4	0.018
-3	0.047
-2	0.119
-1	0.269
0	0.50
1	0.731
2	0.881
3	0.952
4	0.982
5	0.993
6	0.997
7	0.999

Transforming linear output using the sigmoid function

The following equation represents the linear component of a logistic regression model:

$$z = b + w_1x_1 + w_2x_2 + \dots + w_Nx_N$$

where:

z is the output of the linear equation, also called the [log odds](#).

b is the bias.

The w values are the model's learned weights.

The x values are the feature values for a particular example.

To obtain the logistic regression prediction, the z value is then passed to the sigmoid function, yielding a value (a probability) between 0 and 1:

$$y' = \frac{1}{1 + e^{-z}}$$

where:

y' is the output of the logistic regression model.

e is [Euler's number](#): a mathematical constant ≈ 2.71828 .

z is the linear output (as calculated in the preceding equation).

Click here to learn more about log-odds

In the equation $z = b + w_1x_1 + w_2x_2 + \dots + w_Nx_N$, z is referred to as the *log-odds* because if you start with the following sigmoid function (where y is the output of a logistic regression model, representing a probability):

$$y = \frac{1}{1 + e^{-z}}$$

And then solve for z :

$$z = \ln\left(\frac{y}{1 - y}\right)$$

Then z is defined as the natural logarithm of the ratio of the probabilities of the two possible outcomes: y and $1 - y$.

Figure 2 illustrates how linear output is transformed to logistic regression output using these calculations.

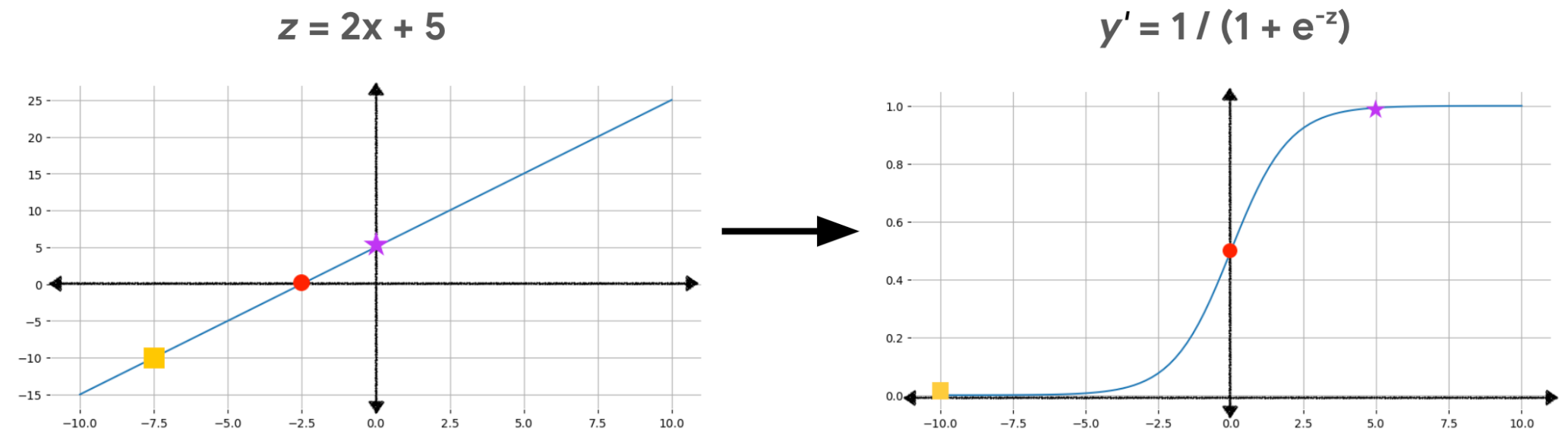


Figure 2. Left: graph of the linear function $z = 2x + 5$, with three points highlighted. Right: Sigmoid curve with the same three points highlighted after being transformed by the sigmoid function.

In Figure 2, a linear equation becomes input to the sigmoid function, which bends the straight line into an s-shape. Notice that the linear equation can output very big or very small values of z , but the output of the sigmoid function, y' , is always between 0 and 1, exclusive. For example, the yellow square on the left graph has a z value of -10 , but the sigmoid function in the right graph maps that -10 into a y' value of 0.00004 .

Exercise: Check your understanding

A logistic regression model with three features has the following bias and weights:

$$\begin{aligned} b &= 1 \\ w_1 &= 2 \\ w_2 &= -1 \\ w_3 &= 5 \end{aligned}$$

Given the following input values:

$$\begin{aligned} x_1 &= 0 \\ x_2 &= 10 \\ x_3 &= 2 \end{aligned}$$

Answer the following two questions.

1. What is the value of z for these input values?

- 1
- 0
- 0.731
- 1

Correct! The linear equation defined by the weights and bias is $z = 1 + 2x_1 - x_2 + 5x_3$. Plugging the input values into the equation produces $z = 1 + (2)(0) - (10) + (5)(2) = 1$

Correct answer.

2. What is the logistic regression prediction for these input values?

- 0.268
- 0.5
- 0.731

As calculated in #1 above, the log-odds for the input values is 1. Plugging that value for z into the sigmoid function:

$$y = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-1}} = \frac{1}{1 + 0.367} = \frac{1}{1.367} = 0.731$$

Correct answer.

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